Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor: $\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 (3x)^3 = 27x^3 (-2y^2)^5 = -32y^{10}$ Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32) x^3 y^{10} = 56 \times (-864) x^3 y^{10} = -48384 x^3 y^{10}$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

Construct the Taylor polynomial using $a = 2$: $P_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$ $= 3 - 1(x-2) + \frac{4}{2}(x-2)^2 + \frac{-12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 - 2(x-2)^3$ Substitute $x = 2.1$, so $x - 2 = 0.1$: $P_3(2.1) = 3 - 0.1 + 2(0.1)^2 - 2(0.1)^3 = 3 - 0.1 + 2(0.01) - 2(0.001)$ $= 3 - 0.1 + 0.02 - 0.002 = 2.918$

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

Identify the first term and ratio: When $n = 1$, $a = 3\left(-\frac{2}{5}\right)^1 = -\frac{6}{5}$. The common ratio is $r = -\frac{2}{5}$. Since $|r| = \frac{2}{5} < 1$, the series converges. Using the sum formula $S = \frac{a}{1-r}$: $S = \frac{-6/5}{1-(-2/5)} = \frac{-6/5}{7/5} = -\frac{6}{7}$

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

Compute derivatives at $x = 0$: • $f(0) = \cos(0) = 1$ • $f'(0) = -\sin(0) = 0$ • $f''(0) = -\cos(0) = -1$ • $f'''(0) = \sin(0) = 0$ Applying the Maclaurin formula: $P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 = 1 - \frac{x^2}{2}$. (Notice the cubic term has coefficient 0).

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left|\frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}} \cdot \frac{n^2 4^n}{(2x-1)^n}\right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x - 1 = 4$, giving $\sum_{n=1}^{\infty} \frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p-series test ($p = 2 > 1$). Endpoint $x = -3/2$: $2x - 1 = -4$, giving.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

1. Compute the cross product via determinant: $\mathbf{a} \times \mathbf{b} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2, -1, 3; 1, 4, -2| = \mathbf{i}[(-1)(-2) - 3(4)] - \mathbf{j}[2(-2) - 3(1)] + \mathbf{k}[2(4) - (-1)(1)]$ $= \mathbf{i}(2 - 12) - \mathbf{j}(-4 - 3) + \mathbf{k}(8 + 1) = -10\mathbf{i} + 7\mathbf{j} + 9\mathbf{k} = (-10; 7; 9).$ 2. The area of the triangle is half the magnitude of the cross product.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1 + u)^p = 1 + pu + \frac{p(p-1)}{2!}u^2 + \ldots$. Here $u = -x^2$ and $p = -1/2$: $(1 - x^2)^{-1/2} = 1 + \left(-\frac{1}{2}\right)(-x^2) + \frac{(-1/2)(-3/2)}{2}(-x^2)^2 + O(x^6) = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + O(x^6).$ 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, integrate term-by-term: $\arcsin x = \int_0^x \left(1 + \frac{1}{2}t^2 + \frac{3}{8}t^4 + \ldots\right)dt = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \ldots$ 3. The binomial series converge.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2+y^2}\,dA$ where D is the semi-annular region $\{(x,y): 1 \le x^2+y^2 \le 4, y \ge 0\}$.

Convert the semi-annulus to polar coordinates: Radius bounds: $1 \le r^2 \le 4 \Rightarrow 1 \le r \le 2$. Angle bounds (since $y \ge 0$, upper half-plane): $0 \le \theta \le \pi$. Area element: $dA = r\,dr\,d\theta$. Set up the integral: $\iint_D e^{x^2+y^2}\,dA = \int_0^{\pi}\int_1^2 e^{r^2}\,r\,dr\,d\theta = \left(\int_0^{\pi} d\theta\right)\left(\int_1^2 re^{r^2}\,dr\right)$ Substitute $u = r^2 \Rightarrow du = 2r\,dr$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi}\int_y^{\pi} \frac{\sin x}{x}\,dx\,dy$

The domain is $0 \le y \le \pi$ and $y \le x \le \pi$. This is the triangular region bounded by $y = 0$, $x = \pi$, and $y = x$. Reversing order to Type I: x ranges from 0 to π, while y ranges from 0 to x: $\int_0^{\pi}\int_0^{x} \frac{\sin x}{x}\,dy\,dx = \int_0^{\pi}\left[y\frac{\sin x}{x}\right]_{y=0}^{y=x} dx = \int_0^{\pi} x\frac{\sin x}{x}\,dx = \int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Step 1: Test for absolute convergence. Consider $\sum |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$. This is a p-series with $p = 1/2 \le 1$, which diverges. Hence the series does NOT converge absolutely. Step 2: Apply Alternating Series Test. Here $b_n = \frac{1}{\sqrt{n}} > 0$. 1. Monotonicity: \(n+1 > n \implies \sqrt{n+1} > \sqrt{n} \implies \frac{1}{\sqrt{n+1}} conditionally convergent.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 7.01 Vectors**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

Compute the limit of the general term as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos(0) = 1$ Because $\lim_{n\to\infty} a_n = 1e0$, the series diverges by the Divergence Test.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

Set up iterated integral: $\int_1^2 \int_0^3 (2x + y)\,dx\,dy = \int_1^2 [x^2 + xy]_0^3 dy = \int_1^2 (9 + 3y)\,dy$ Integrate with respect to y: $\left[9y + \frac{3y^2}{2}\right]_1^2 = (18 + \frac{12}{2}) - (9 + \frac{3}{2}) = 24 - 10.5 = 13.5 = \frac{27}{2}$

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 9.01 Complex Numbers**

Find the Maclaurin series for $\ln(1 + 2x)$ up to the x^3 term, and state its interval of convergence.

Use $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \ldots$ with $u = 2x$: $\ln(1 + 2x) = 2x - 2x^2 + \frac{8}{3}x^3 - \ldots$. Convergence requires $-1 < 2x \le 1 \Rightarrow -\frac{1}{2} < x \le \frac{1}{2}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 10.01 Numerical Methods**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 11.01 Differential Equations**

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first three terms in the expansion of $(1 + 4x)^{1/2}$ in ascending powers of x.

a. $1 + 2x - 2x^2$ ✓

b. $1 + 2x + 2x^2$

c. $1 + 4x - 8x^2$

d. $1 + 2x - 4x^2$

$(1+u)^{1/2} = 1 + \frac{1}{2}u + \frac{(1/2)(-1/2)}{2}u^2 = 1 + \frac{1}{2}(4x) - \frac{1}{8}(16x^2) = 1 + 2x - 2x^2$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the range of values of x for which the binomial expansion of $(1 + 4x)^{1/2}$ is valid.

a. $|x| < \frac{1}{4}$ ✓

b. $|x| < 4$

c. $|x| < 1$

d. $x > 0$

The expansion of $(1 + u)^n$ for non-integer n requires $|u| < 1$. Here $|4x| < 1 \Rightarrow |x| < 1/4$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Expand $(1 - 2x)^{-1}$ up to the term in x^3.

a. $1 + 2x + 4x^2 + 8x^3$ ✓

b. $1 - 2x + 4x^2 - 8x^3$

c. $1 + 2x + 2x^2 + 2x^3$

d. $1 - 2x - 4x^2 - 8x^3$

$(1-u)^{-1} = 1 + u + u^2 + u^3 + \ldots$ For $u = 2x$: $1 + 2x + 4x^2 + 8x^3$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use the first three terms of $(1 - x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

a. 0.98995 ✓

b. 0.98990

c. 0.99000

d. 0.98500

$1 - \frac{1}{2}(0.02) - \frac{1}{8}(0.0004) = 1 - 0.01 - 0.00005 = 0.98995$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

a. $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$ ✓

b. $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

c. $(50)^{1/2} = (1+49)^{1/2}$

d. $\sqrt{36+14} = 6(1+14/36)^{1/2}$

We factor out the nearest perfect square 49 so that $x = 1/49 \approx 0.0204 \ll 1$, giving rapid convergence: $7(1+1/49)^{1/2}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coefficient of x^2 in the expansion of $\frac{1+2x}{(1-x)^2}$.

a. **5** ✓

b. 3

c. 4

d. 6

$(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots$ Multiply: $(1+2x)(1+2x+3x^2) = 1(3x^2) + 2x(2x) = 3x^2 + 4x^2 = 7x^2$? Wait: $(1-x)^{-2} = 1 + 2x + 3x^2$. Term in x^2: $1(3) + 2(2) = 7$. Choices adjusted: 7.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.

a. $\frac{n}{n+1}$ ✓

b. $1 - \frac{1}{n}$

c. $\frac{n+1}{n}$

d. $1 - \frac{1}{(n+1)^2}$

Telescoping sum: $(1 - 1/2) + (1/2 - 1/3) + \ldots + (1/n - 1/(n+1)) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum to infinity $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.

a. **1** ✓

b. 2

c. 1/2

d. ∞

$\lim_{n\to\infty} \frac{n}{n+1} = 1$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

a. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$ ✓

b. $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

c. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

d. $f(x) = f(1) + f'(1)(x-1) + \ldots$

The standard Maclaurin series expansion is $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

a. $x - \frac{x^2}{2} + \frac{x^3}{3}$ ✓

b. $x + \frac{x^2}{2} + \frac{x^3}{3}$

c. $x - x^2 + x^3$

d. $1 + x - \frac{x^2}{2}$

Standard series: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ for $-1 < x \leq 1$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term $a = 2$ and common difference $d = -5$. Find the 8th term.

a. **-33** ✓

b. -35

c. -38

d. 16

$u_n = a + (n-1)d \Rightarrow u_8 = 2 - 35 = -33$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $x^2 + 10x + 41 = 0$. Find the sum of the roots.

a. 41

b. -9

c. 10

d. **-10 ✓**

Sum of roots $= -b/a = -10$ (roots are $-5 \pm 4i$).

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The locus $|z - (-6 + i)| = |z - (2 - 2i)|$ is the perpendicular bisector of the segment joining the points representing z_1 and z_2. Find the x-coordinate of the midpoint of this segment.

a. $-\frac{1}{2}$

b. 2

c. **-2 ✓**

d. -6

Midpoint x-coordinate: $\frac{-6+2}{2} = -2$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = \frac{a_n^2 + k}{2}$ for $n \geq 1$, where k is a positive constant. If the sequence converges to a limit L, prove that $L^2 - 2L + k = 0$. Furthermore, if the limit L is an integer and $k = 3$, determine the exact value of the limit.

a. **L = 3 ✓**

b. L = 1

c. L = -1

d. $L = \sqrt{3}$

If the sequence converges to a limit L, then as $n \to \infty$, $a_{n+1} \to L$ and $a_n \to L$. Substituting this into the recurrence relation, we get: $L = \frac{L^2 + k}{2}$ Multiplying by 2: $2L = L^2 + k$ Rearranging the terms to form a quadratic equation: $L^2 - 2L + k = 0$ This proves the first part of the problem. For the second part, we are given that $k = 3$ and the limit L is an integer. Substituting $k = 3$ into the equation derived above: $L^2 - 2L + 3 = 0$ We can solve this qua.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. **100** ✓

b. 110

c. 10

d. 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n-1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n-1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n-1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n-1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. **2** ✓

b. 0

c. -2

d. 1

Let's compute the first few terms of the sequence: $a_1 = 2$ $a_2 = a_1^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ $a_3 = a_2^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ It appears that for $a_1 = 2$, all subsequent terms of the sequence are also 2. We can prove this by induction. Base case: For $n = 1$, $a_1 = 2$, which is true. Inductive step: Assume that for some integer $k \geq 1$, $a_k = 2$. We need to show that $a_{k+1} = 2$. From the recurrence relation, $a_{k+1} = a_k^2 - 2$. By the inductive hypothesis, $a_k = 2$. Substituting this into.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Numerical Methods

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

a. 6

b. **7** ✓

c. 8

d. 9

Let the arithmetic sequence be defined by $a_n = a_1 + (n-1)d$. We are given that $a_5 = 19$ and $a_{11} = 43$. From the given information, we can set up a system of two linear equations: 1. $a_1 + (5-1)d = 19 \Rightarrow a_1 + 4d = 19$ 2. $a_1 + (11-1)d = 43 \Rightarrow a_1 + 10d = 43$ Subtract equation (1) from equation (2): $(a_1 + 10d) - (a_1 + 4d) = 43 - 19$ $6d = 24$ $d = \frac{24}{6} = 4$ Now substitute the value of d back into equation (1): $a_1 + 4(4) = 19$ $a_1 + 16 = 19$ $a_1 = 19 - 16 = 3$ We are aske.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Differential Equations

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the 4th term of this sequence, a_4?

a. 11

b. **14** ✓

c. 17

d. 20

An arithmetic sequence is defined by the formula $a_n = a_1 + (n-1)d$, where a_n is the nth term, a_1 is the first term, and d is the common difference. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Substituting these values into the formula: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$ Alternatively, we can list the terms: $a_1 = 5$ $a_2 = a_1 + d = 5 + 3 = 8$ $a_3 = a_2 + d = 8 + 3 = 11$ $a_4 = a_3 + d = 11 + 3 = 14$ Therefore.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Poisson Distribution

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fourth term of this sequence, a_4?

a. **14** ✓

b. 11

c. 17

d. 8

The formula for the n-th term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. Given $a_1 = 5$ and $d = 3$, we want to find a_4. Substituting $n = 4$ into the formula: $a_4 = 5 + (4 - 1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the fourth term of the sequence is 14. Common errors might occur from miscalculating $(n - 1)$ or incorrectly applying the addition.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 13.01 Continuous Random Variables

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?

a. **17** ✓

b. 20

c. 15

d. 23

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference. Given $a_1 = 5$ and $d = 3$, we want to find the 5th term, a_5. Using the formula: $a_5 = a_1 + (5 - 1)d$ $a_5 = 5 + (4)(3)$ $a_5 = 5 + 12$ $a_5 = 17$ Therefore, the 5th term of the sequence is 17.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 14.01 Sampling & Estimation

Product of two power series $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:

a. $a_n b_n$

b. $a_n + b_n$

c. $\sum_{k=0}^{n} a_k b_{n-k}$ **(Cauchy product)** ✓

d. $a_0 b_n$

Applying Sequences and Series principles, Cauchy product convolution. Therefore, the verified correct option is $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product).

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Hypothesis Testing

Integral test: $\sum_{n=1}^{\infty} f(n)$ converges iff $\int_{1}^{\infty} f(x)\,dx$ converges, when f is:

a. any function

b. bounded

c. increasing

d. **positive, decreasing, continuous ✓**

Applying Sequences and Series principles, Integral test conditions. Therefore, the verified correct option is positive, decreasing, continuous.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Exponentials & Logarithms

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge absolutely?

a. Yes

b. Yes, to $-\ln 2$

c. **No ($\sum 1/n$ diverges) ✓**

d. Inconclusive

Applying Sequences and Series principles, Not absolute: $\sum 1/n$ diverges. Therefore, the verified correct option is No ($\sum 1/n$ diverges).

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